

Box 14.1 Causes of Hypotonic Polyuria

Arginine Vasopressin Deficiency (Central Diabetes Insipidus)

- Congenital (congenital nephrogenesis, autosomal dominant, or -argine vasopressin (AVP) neurexin gene mutations)
- Drug- or toxin-induced (lithium, diuretic, fluorination, male ven-erally)
- Dissecting (hypocystitis, vasculitis)
- Neoplastic (paraneoplastic, germinoma, lymphoma, leuke-mia, meningioma, pituitary tumor, metastasis)
- Infectious (meningitis, tuberculosis, cryptococcosis)
- Inflammatory (sarcoidosis, lymphocytic, HIV-associated, encephalitis)
- Thalamic (neurosurgery, deceleration injury)
- Vascular (cerebral hemorrhage or infarction, brain death)
- Idiopathic

Osmoreceptor Dysfunction

- Genetic (X-linked recessive, AVP-V₁ receptor gene muta-tions; autosomal recessive or dominant, aquaporin-2 water channel gene mutations)
- Vascular (anterior communicating artery aneurysm or ligation, aneurysmal/hemorrhagic)
- Other (hypophysitis, ventricular or suprasellar cyst, trauma, degenerative disease)
- Idiopathic

Increased AVP Metabolism

A rare Vasopressin Resistance (Nephrogenic Diabetes Insipidus)

- Genetic (X-linked recessive, AVP-V₁ receptor gene muta-tions; autosomal recessive or dominant, aquaporin-2 water channel gene mutations)
- Drug-induced (demeclocycline, lithium, captopril, methoxyfu-riol)
- Hypokalemia
- Hypomagnesemia
- Infiltrating lesions (sarcoidosis, amyloidosis)
- Vascular (stroke of anterior)
- Mechanical (polydipsia, kidney mass, bilateral ureteral ob-struction)
- Solute diuresis (glucose, mannitol, sodium, radiocontrast dye)
- Idiopathic

Primary Polydipsia

- Psychogenic (schizophrenia, obsessive-compulsive behavior)
- Dipsogenic (downward resetting of thirst threshold, idiopathic, or similar lesions as with AVP-D)

Box_14_1

Box 14.2 Therapies for the Treatment of AVP-D and AVP-R

Water

- Antidiuretic agents
- Arginine vasopressin (Pitressin)
- 1-Deamino-8-D-arginine vasopressin (desmopressin; DDAVP)
- Antidiuresis-enhancing agents
- Chlorpropamide
- Prostaglandin synthase inhibitors (e.g., indomethacin, ibu-phen, and tolmetin)
- Natriuretic agents
- Thiazide diuretics
- Amiloride

Box_14_2

Box 14.3 Disorders Associated With Syndrome of Inappropriate Antidiuresis

Tumors

- Pituitary (adenomatous, bronchogenic, carcinoma, metastati-cally)
- Thyroidal
- Meningeal (adenocarcinoma, pancreatic carcinoma, metast-atized, leukemia)

Central Nervous System Disorders

- Mass lesions (tumor, hemorrhage, abscess, subarachnoid hemor-rhage)
- Inflammatory diseases (encephalitis, meningitis, systemic lupus erythematosus, acute intermittent porphyria, multiple sclero-sis)
- Depressive/temperamental diseases (Guller-Bard syndrome, apical cord lesions)
- Metastatic (radiation-induced hemorrhage, head trauma, acute psychosis, sodium meninges, pituitary stalk section, trans-sphenoidal adenectomy, hydrocephalus)

Drug/Medication Related

- Stimulated release of AVP (cocaine, phenothiazines, tricyclics)
- Direct oral effects or potentiation of AVP (antidiuretic effects: (desmopressin, oxytocin, prostaglandin synthase inhibitors)
- Mixed or unrelated actions (ACE inhibitors, carbamazepine, and oxcarbazepine, chlorpropamide, dobutamine, diazepam, cyclo-phosphamide, 5,6-methylenedioxymethamphetamine (ecsta-sy), omegazine, serotonin reuptake inhibitors, vincristine)

Pulmonary

- Infectious (subacute, acute bacterial and viral pneumonia, aspergillosis, emphysema)
- Mechanical (ventilatory causes (acute respiratory failure, COPD, positive-pressure ventilation)

Other Causes

- Acquired immunodeficiency syndrome (AIDS) and AIDS-related complex
- Prolonged strenuous exercise (marathon, triathlon, ultra-marathon, hot weather hiking)
- Severe alcoholism
- Idiopathic

ACE, Angiotensin-converting enzyme; AVP, arginine vasopressin; COPD, chronic obstructive pulmonary disease.

Box_14_3

Box 14.4 General Recommendations for Using Fluid Restriction and Predictors of Its Increased Likelihood of Failure

General Recommendations

- Restrict all intake that is consumed by drinking, not just wa-ter.
- Aim for a fluid restriction that is 500 mL/day below the 24-hour urine volume.
- Do not restrict sodium or protein intake unless indicated.

Predictors of Likely Failure of Fluid Restriction

- High urine osmolality (>500 mOsm/kg H₂O)
- Sum of urine Na⁺ and K⁺ concentrations exceeds serum Na⁺ concentration
- 24-hour urine volume < 1500 mL/day
- Increase in serum Na⁺ sodium concentration < 2 mmol/L/day in 24 hours on fluid restriction ≤ 1 L/day

From Verbalis JG, Goldsmith SR, Greenberg A, et al. Diagnosis, evaluation, and treatment of hyponatremia: expert panel recommendations. *Am J Med.* 2013;126(Suppl 1):S1–S42.

Box_14_4

Box 14.5 Factors Increasing Risk of Osmotic Demyelination Syndrome^a

- Serum sodium concentration ≤105 mmol/L
- Hypokalemia^b
- Alcoholism^b
- Malnutrition^b
- Advanced liver disease^b

^aRequiring slower correction of chronic hyponatremia.

^bUnlike the rate of increase in serum [Na⁺], neither the precise level of the serum potassium concentration nor the degree of alcoholism, malnutrition, or liver disease that alters the brain's tolerance to an acute osmotic stress have been rigorously defined.

From Verbalis JG, Goldsmith SR, Greenberg A, et al. Diagnosis, eval-uation, and treatment of hyponatremia: expert panel recommen-dations. *Am J Med.* 2013;126(Suppl 1):S1–S42.

Box_14_5

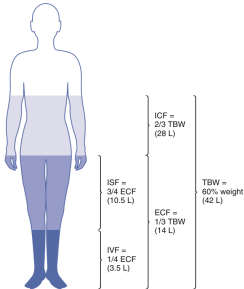


Fig. 14.1 Schematic representation of body fluid compartments in humans. The shaded areas depict the approximate size of each compartment as a function of body weight. The numbers indicate the relative sizes of the various fluid compartments and the approximate absolute volumes of the compartments (in liters) in a 70-kg adult. ECF, Extracellular fluid; ICF, intracellular fluid; ISF, interstitial fluid; IVF, intravascular fluid; TBW, total body water. (From Verbalis JG. Body water and osmolality. In: Wilkinson D, Jamison R, eds. *Textbook of Nephrology*. London: Chapman & Hall, 1997:39–44.)

Fig_14_1

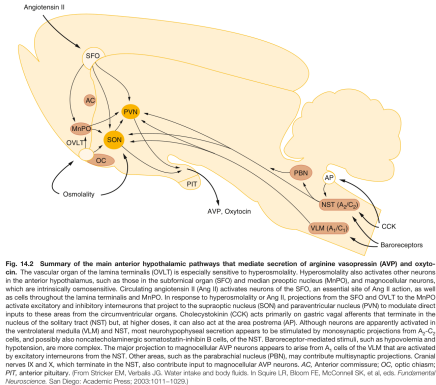


Fig. 14.2 Summary of the main anterior hypothalamic pathways that mediate secretion of arginine vasopressin (AVP) and oxytocin. The supraoptic nucleus (SON) is especially sensitive to hyperosmolality. Hyperosmolality also activates other neurons in the anterior hypothalamus, such as those in the additional optic tract (AOT) and median preoptic nucleus (MPO). The supraoptic nucleus, which are intrinsically osmosensitive. Circulating angiotensin II (Ang II) activates neurons of the SON, an essential site of Ang II action, as well as cells throughout the lamina terminalis (MPO). In response to hyperosmolality of Ang II, projections from the SON and PVN to the MPO activate excitatory and inhibitory interneurons that project to the supraoptic nucleus (SON) and paraventricular nucleus (PVN) to modulate direct inputs to these areas from the circumventricular organs. Cholecystokinin (CCK) acts primarily on gastric vagal afferents that terminate in the nucleus of the solitary tract (NTS) but, at higher doses, it can also act at the area postrema (AP). Although neurons are apparently activated in the ventrolateral medulla (VLM) and NTS, most neurohypophyseal secretion appears to be stimulated by monosynaptic projections from A₁C₁ cells, and possibly also nociceptin/hemarginin somatostatin-inhibitory (N) cells, of the NTS. Baroreceptor-mediated stimuli, such as hypotension and hypernatremia, are more complex. The major projection to magnocellular AVP neurons appears to arise from A₁ cells of the VLM that are activated by excitatory interneurons from the NTS. Other areas, such as the parabrachial nucleus (PBN), may contribute multisynaptic projections. Central nervous system (CNS) and X, which terminates in the NTS, also contribute input to magnocellular AVP neurons. AC, anterior commissure; OC, optic chiasm; PVT, anterior pituitary. (From Verbalis JG, Goldsmith SR, Greenberg A, et al. Diagnosis, evaluation, and treatment of hyponatremia: expert panel recommendations. *Am J Med.* 2013;126(Suppl 1):S1–S42.)

Fig_14_2

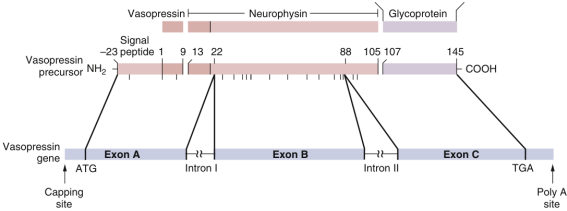


Fig. 14.3 The arginine vasopressin (AVP) gene and its protein products. The three exons encode a 145-amino acid prohormone with an NH₂-terminal signal peptide. The prohormone is packaged into neurosecretory granules of magnocellular neurons. During axonal transport of the granules from the hypothalamus to the posterior pituitary, enzymatic cleavage of the prohormone generates the final products—AVP, neurophysin, and a COOH-terminal glycoprotein called copeptin. When afferent stimulation depolarizes the AVP-containing neurons, the three products are released into capillaries of the posterior pituitary in equimolar amounts. (Adapted from Richter Z, Schmale H. The structure of the precursor to arginine vasopressin, a model preprohormone. *Prog Brain Res.* 1983;60:227–233.)

Fig_14_3

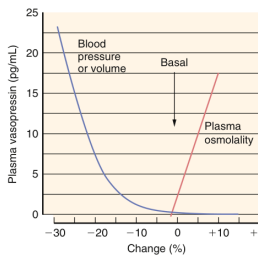


Fig. 14.4 Comparative sensitivity of arginine vasopressin (AVP) secretion in response to increases in plasma osmolality versus decreases in blood volume or blood pressure in human subjects. The arrow indicates the low plasma AVP concentrations found at basal plasma osmolality. Note that AVP secretion is much more sensitive to small changes in blood osmolality than to small changes in volume or pressure. (Adapted from Robertson GL. Posterior pituitary. In: Felig P, Baxter J, Frohman LA, eds. *Endocrinology and Metabolism*. New York: McGraw-Hill, 1986:338–386.)

Fig_14_4

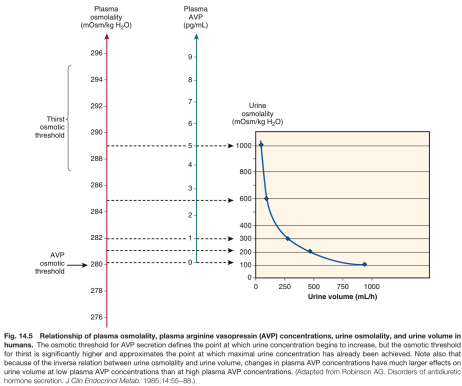


Fig. 14.5 Relationship of plasma osmolality, plasma arginine vasopressin (AVP) concentrations, urine osmolality, and urine volume in humans. The osmotic threshold for AVP secretion defines the point at which urine concentration begins to increase, but the osmotic threshold for thirst is significantly higher and approximates the point at which maximal urine concentration has already been achieved. Note also that because of the inverse relation between urine osmolality and urine volume, changes in plasma AVP concentrations have much larger effects on urine volume at low plasma AVP concentrations than at high plasma AVP concentrations. (Adapted from Robertson AG. Disorders of antidiuretic hormone secretion. J Clin Endocrinol Metab. 1985;14:55-89.)

Fig_14_5

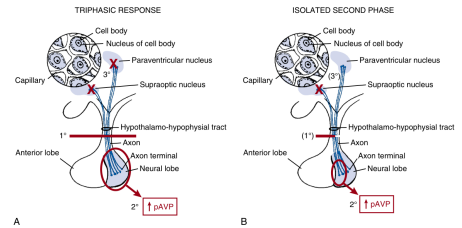
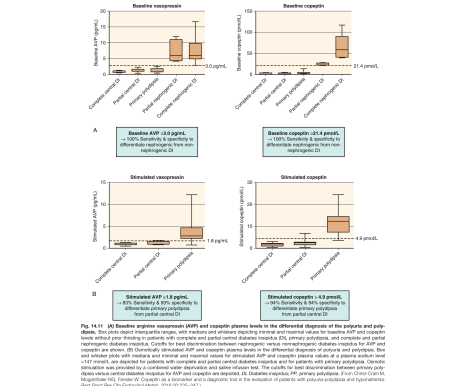


Fig. 14.8 Mechanisms underlying the pathophysiology of the triphasic pattern of arginine vasopressin (AVP-D) and the isolated second phase. (A) In the triphasic response, the first phase of AVP-D is initiated following a partial or complete pituitary stalk section, which severs the connections between the AVP neuronal cell bodies in the hypothalamus and nerve terminals in the posterior pituitary gland, thus preventing stimulated AVP secretion (1 degree). This is followed in several days by the second phase of SIAD, which is caused by uncontrolled release of AVP into the bloodstream from the degenerating nerve terminals in the posterior pituitary (2 degree). After all the AVP stored in the posterior pituitary gland has been released, the third phase of DI returns if more than 80% to 90% of the AVP neuronal cell bodies in the hypothalamus have undergone retrograde degeneration (3 degree). (B) In the isolated second phase, the pituitary stalk is injured, but not completely cut. Although the maximum AVP secretory response will be diminished as a result of the stalk injury, DI will not result if the injury leaves intact at least 10% to 20% of the nerve fibers connecting the AVP neuronal cell bodies in the hypothalamus to the nerve terminals in the posterior pituitary gland (1 degree). However, this is still followed in several days by the second phase of SIAD, which is caused by the uncontrolled release of AVP from the degenerating nerve terminals of the posterior pituitary gland that have been injured or severed (2 degree). Because a smaller portion of the posterior pituitary is denervated, the magnitude of AVP released as the pituitary degenerates will be smaller and of shorter duration than with a complete stalk section. After all the AVP stored in the damaged part of the posterior pituitary gland has been released, the second phase ceases, but clinical DI will not occur if <80% to 90% of the AVP neuronal cell bodies in the hypothalamus undergo retrograde degeneration (3 degree). (From Loh JA, Verbalis JG. Disorders of water and salt metabolism associated with pituitary disease. Endocrinol Metabol Clin North Am. 2008;37:2113-224.)

Fig_14_8



Fig_14_11

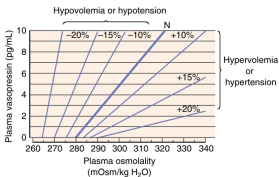


Fig. 14.6 The relationship between the osmolality of plasma and concentration of arginine vasopressin (AVP) in plasma is modulated by blood volume and pressure. The line labeled N shows plasma AVP concentration across a range of plasma osmolalities in an adult with normal intravascular volume (i.e., euvolemic) and normal blood pressure (i.e., normotensive). The lines to the left of N show the relationship between plasma AVP concentration and plasma osmolality in adults whose low intravascular volume (i.e., hypovolemia) or blood pressure (i.e., hypotension) is 10%, 15%, and 20% below normal, respectively. Lines to the right of N indicate volumes and blood pressures 10%, 15%, and 20% above normal, respectively. Note that hemodynamic influences do not disrupt the osmoregulation of AVP but rather raise or lower the set point, and possibly also the sensitivity, of AVP secretion in proportion to the magnitude of the change in blood volume or pressure. (Adapted from Robertson GL, Athar S, Shelton RL. Osmotic control of vasopressin function. In Andreoli TE, Grantham JJ, Rector FC Jr, eds. Disorders in Body Fluid Osmolality. Bethesda, MD: American Physiology Society; 1977:125.)

Fig_14_6

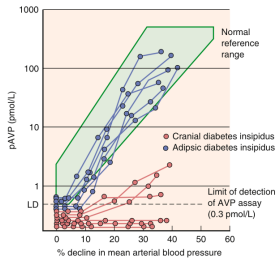


Fig. 14.9 Plasma arginine vasopressin (AVP) responses to arterial hypotension produced by the infusion of trimethaphan in patients with AVP-D (central diabetes insipidus) and osmoreceptor dysfunction (adipsic AVP-D). Normal responses in healthy volunteers are shown by the shaded area. Note that despite absent or markedly blunted AVP responses to hypotension, patients with osmoreceptor dysfunction respond normally to baroreceptor stimulation induced by hypotension. (From Baylis FH, Thompson CL. Diabetes insipidus and hyponatremic syndromes. In Becker KL, ed. Principles and Practice of Endocrinology and Metabolism. Philadelphia: JB Lippincott; 1995:257.)

Fig_14_9

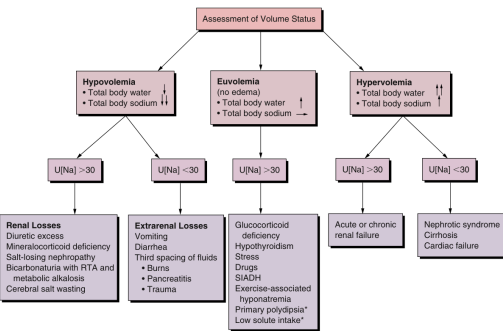


Fig. 14.12 Diagnostic approach to the hyponatremic patient. SIAD, Syndrome of inappropriate antidiuretic hormone secretion; U[Na], urinary [Na⁺]. (Modified from Halperin R, Berl T. Therapy of dysnatremic disorders. In Brady H, Wilcox C, eds. Therapy in Nephrology and Hypertension, Philadelphia: WB Saunders; 1999:256.)

Fig_14_12

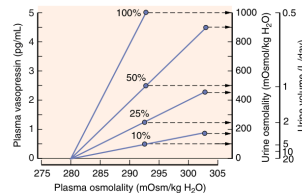


Fig. 14.7 Relationship between plasma arginine vasopressin (AVP) levels, urine osmolality, and plasma osmolality in subjects with normal posterior pituitary function (100%) compared with patients with graded reductions in AVP-secreting neurons (to 50%, 25%, and 10% of normal). Note that the patient with a 50% secretory capacity can achieve only half the plasma AVP level and half the urine osmolality of normal subjects at a plasma osmolality of 293 mOsm/kg H₂O. However, with increasing plasma osmolality, this patient can nonetheless eventually stimulate sufficient AVP secretion to reach a near-maximal urine osmolality. In contrast, patients with more severe degrees of AVP-secreting neuron deficits are unable to reach maximal urine osmolalities at any level of plasma osmolality. (Adapted from Robertson GL. Posterior pituitary. In Felig P, Baxter J, Frohman LA, eds. Endocrinology and Metabolism. New York: McGraw-Hill; 1980:338-389.)

Fig_14_7

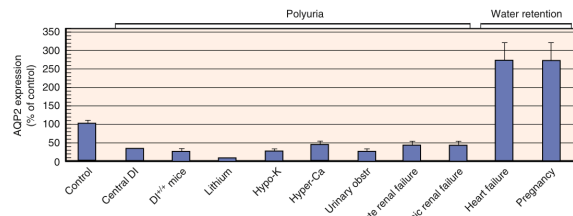


Fig. 14.10 Kidney expression of the water channel aquaporin-2 in various animal models of polyuria and water retention. Note that kidney aquaporin-2 expression is uniformly downregulated relative to levels in controls in all animal models of polyuria but upregulated in animal models of inappropriate antidiuresis. DI⁺, Genetic diabetes insipidus; Hyper-Ca, hypercalcemia; Hypo-K, hypokalemia; Urinary obstr, ureteral obstruction. (From Nielsen S, Kwon TH, Christensen BM, et al. Physiology and pathophysiology of renal aquaporins. J Am Soc Nephrol. 1999;10:647-663.)

Fig_14_10

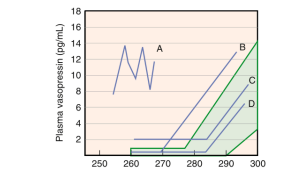
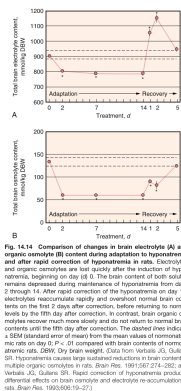


Fig. 14.13 Plasma vasopressin as a function of plasma osmolality during the infusion of hypertonic saline in four groups of patients with the clinical syndrome of inappropriate antidiuresis (SIAD). Type A osmoregulatory defect was associated with large and erratic fluctuations in plasma AVP, which bore no relation to the rise in plasma osmolality. Type B osmoregulatory defect was consistent with a downward resetting of the osmostat for AVP secretion. Type C osmoregulatory defect was associated with elevated plasma AVP initially, which did not change during the infusion of hypertonic saline until plasma osmolality reached the normal range, after which plasma AVP began to rise appropriately, indicating a normally functioning osmoreceptor mechanism. Type D osmoregulatory defect was associated with normal stimulation and suppression of plasma AVP secretion, despite a marked inability to excrete a water load. Shaded area, Range of normal values. (From Zerle R, Shipes L, Robertson G. Vasopressin function in the syndrome of inappropriate antidiuresis. Annu Rev Med. 1980;31:315-327.)

Fig_14_13



Fig_14_14

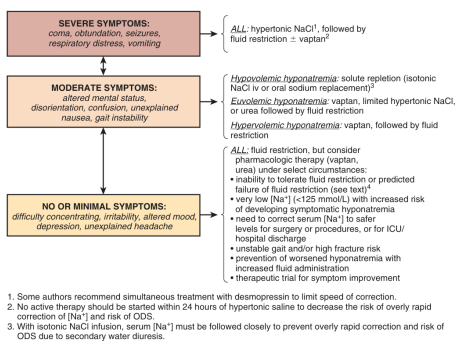


Fig. 14.15 Algorithm for the treatment of patients with euolemic hyponatremia based on their presenting symptoms. The arrows between the symptom boxes indicate movement of patients between different symptom levels; ALL, All types of hyponatremia; ICU, intensive care unit; iv, intravenous; ODS, osmotic demyelination syndrome (Modified from Verbalis JG. Emergency management of acute and chronic hyponatremia. In: Matlin G, ed. *Endocrine and Metabolic Emergencies*. Washington, DC: Endocrine Press; 2014:359.)

Fig_14_15

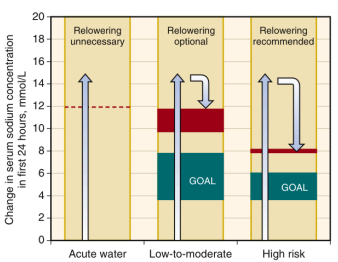


Fig. 14.16 Recommended goals (green) and limits (red) for correction of hyponatremia based on the risk of producing osmotic demyelination syndrome (ODS). Also shown are recommendations for relowering of the serum [Na⁺] to goals for patients presenting with serum [Na⁺] <120 mmol/L who exceed the recommended limits of correction in the first 24 hours. (From Verbalis JG, Goldsmith SR, Greenberg A, et al. Diagnosis, evaluation, and treatment of hyponatremia: expert panel recommendations. *Am J Med*. 2013;126(Suppl 1):S1–S42.)

Fig_14_16

Table 14.1 Relationship Between Serum Tonicity and Sodium Concentration in the Presence of Other Substances			
Condition or Substance	Serum Osmolality	Serum Tonicity	Serum [Na ⁺]
Hyperglycemia	↑	↑	↓
Mannitol, maltose, glycine	↑	↑	↓
Azotemia (high blood urea)	↑	↔	↔
Ingestion of ethanol, methanol, ethylene glycol	↑	↔	↔
Elevated serum lipid or protein	↔	↔	↓

↑, Increased; ↓, decreased; ↔, unchanged.

Table_14_1

Table 14.2 Classification of Hyponatremia According to Severity of Presenting Symptoms			
Severity	Serum Sodium Level	Neurologic Symptoms	Typical Duration of Hyponatremia
Severe	Generally <125 mmol/L	Vomiting, seizures, obtundation, respiratory distress, coma	Acute (<24-48 hours)
Moderate	Generally <130 mmol/L	Nausea, confusion, disorientation, altered mental status, unstable gait, falls	Intermediate or chronic (>24-48 hours)
Mild	<135 mmol/L	Headache, irritability, difficulty concentrating, altered mood, depression	Chronic (several days to many weeks or months)

Table_14_2

Table 14.3 Efficacy of Different Treatments Used as Monotherapy for Syndrome of Inappropriate Antidiuretic Hormone Secretion from the Hyponatremia Registry (%)			
Treatment	δ [Na ⁺] ≥5 mmol/L	[Na ⁺] ≥ 130 mmol/L	[Na ⁺] ≥ 135 mmol/L
No treatment (n = 168)	41	45	20
Fluid restriction (n = 625)	44	29	10
Isotonic saline (n = 384)	36	20	4
Tolvaptan (n = 183)	78	74	40
3% NaCl (n = 78)	60	25	13

From Verbalis JG, Greenberg A, Burst V, et al. Diagnosing and treating the syndrome of inappropriate antidiuretic hormone secretion. *Am J Med*. 2016;129:537.e9–537.e23.

Table_14_3